



PROPOSAL · SUBMITTED TO GXS BANK · ADAPTIVE RISK
MODELLING CHALLENGE

AURA

Adaptive Unified Risk Architecture

ONE MODEL OF YOU. EVERY RISK YOU'LL EVER BE.

AURA is the first risk system where the fraud model and the credit model are the **same model** — observed through different time telescopes — with a **living immune system** that attacks itself 24/7 so real fraudsters discover the blind spots second.

20–30%

CREDIT LOSS REDUCTION

40–50%

FRAUD LOSS REDUCTION

~\$50–80M

ANNUAL P&L IMPACT

PREPARED BY
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01 Executive Summary

GXS Bank has posed the defining risk-modelling question of this decade: how do we unify credit and fraud assessment into a single, continuously learning framework? **AURA — Adaptive Unified Risk Architecture** is our answer. It is the first production-grade system that does not integrate credit and fraud; it recognises they were never separate.

Credit default and fraud are the same phenomenon — the divergence between the customer you underwrote and the customer revealed by their behaviour — observed at different time horizons through different sensors. Today they sit in separate teams, models, and data, with handoff points that are exactly where losses compound. AURA replaces this with one model, one score, one truth — an adaptive architecture that fuses a continuously-learning risk engine with a living immune system, attacking itself 24/7 so real fraudsters discover the blind spots second. For a mid-sized bank with a US\$50 billion portfolio, AURA is projected to deliver 20–30% reduction in credit losses, 40–50% reduction in fraud losses, and approximately US\$50–80 million in annual P&L impact within three years.

02 The Problem: Silos That Bleed

For seven decades, banking has treated credit risk and fraud risk as two distinct sciences. Different teams, models, data, and KPIs. The mathematical reality is that they are the same phenomenon observed at different time horizons through different sensors — and the handoff points between them are precisely where losses compound today.

A bust-out fraud is a credit default compressed into thirty days. A strategic default is a fraud that took eighteen months to reveal itself. When a fraudster becomes indistinguishable from a credit customer until it is too late, or when a distressed credit customer tips into first-party fraud at the moment of crisis, current architectures catch it too late — every single time.

The problem is not to integrate two models. The problem is to stop pretending they were ever separate.

FIGURE 1. The Unified Risk Reframe

Today's Label	What It Actually Is
Credit default	A fraud that took 18 months to reveal itself
Bust-out fraud	A credit default compressed into 30 days
Application fraud	A credit model that was lied to at t=0
Transaction fraud	A creditworthiness event occurring in 30 seconds
Strategic default	A credit customer who transitioned into first-party fraud

03 The Solution: AURA Architecture

AURA is built on eight tightly-coupled layers. The first five form the **Risk Continuum Spine** — one model, one score, one truth, observed through different time telescopes. Layers 6–8 form the **Adaptive Immune System** — an architecture that fights the arms race against itself 24/7 and

converts risk detection from a verdict into a dialogue.

FIGURE 2. The Eight-Layer AURA Architecture

Layer	Function	Why It Matters
1. Unified Risk Embedding	256-dim global latent space for every customer, account, device.	Borrower becomes a moving point in risk space, not a static score.
2. Personal Doppelgänger	Per-customer self-supervised generative twin. Streams residuals where reality diverges.	Detects change vs the customer's own past, not the population. Strictly more sensitive for bust-out.
3. Twin Strands (Capacity × Intent)	Two decoupled readouts: capacity to repay, intent to repay.	Removes the conflation that breaks today's models: low income ≠ low intent.
4. Time-Telescope Heads	Projection heads at spike (sec) / shift (days) / drift (months) horizons.	Fraud team and credit team look at the same customer through different telescopes.
5. Counterfactual Replay Loop	For every decision, simulate N counterfactual trajectories. Train via doubly-robust causal estimation.	The RLHF moment for risk: from predicting outcomes to optimising decisions.
6. The Arena	Adversarial agents (synthetic fraud rings, deepfakes) attack the live system in shadow. Every exploit becomes training signal.	Adaptive = live immune response, not quarterly retraining.
7. Risk Field	Knowledge graph propagating divergence across devices, employers, merchants, geographies.	A fraud ring and a sector layoff are both contagion — same phenomenon, different velocities.
8. Action Layer	Graduated response: nudge → friction → reprice → interdict.	Converts risk system from loss-blocker into cure-generator.

Two elements deserve special attention. **Layer 2 (Doppelgänger)** leverages the 99% of behavioural signal both silos currently throw away — a self-supervised generative twin per customer that constantly predicts their next state and streams the residuals. **Layer 6 (Arena)** is what "adaptive" actually means: a live immune response where adversarial agents attack the system 24/7 so real fraudsters — who now use generative AI themselves — discover the blind spots second.

04 Why Now & Market Opportunity

Three forces converge to make AURA possible in 2026. First, the maturation of causal-inference tooling (EconML, DoWhy, doubly-robust estimators) makes counterfactual training operationalisable at production scale. Second, streaming sequence models make millions of live generative twins computationally feasible. Third, fraudsters now use generative AI — you cannot fight generative with static; only generative with generative.

US\$84T

WEALTH TRANSFER TO
DIGITALLY-NATIVE HNW
CLIENTS BY 2045

~\$487B

GLOBAL ANNUAL FRAUD
LOSSES ACROSS FINANCIAL
SERVICES

10×

FASTER MODEL DEPLOYMENT
VS. SILOED MRM CYCLES

Why GXS specifically. GXS serves the thin-file customers the bureau cannot price — gig workers, young earners, the underbanked. When there is no credit history, behaviour is the only signal, and AURA is a behaviour-first engine. GXS's digital-native data exhaust is exactly the fuel this system needs. This is not a bolt-on; it is GXS's structural advantage weaponised.

05 See It Work

Mei, 34. Real person, real documents. Her doppelgänger detects: salary deposits landing later each month, three new digital credit accounts, top-ups to an unfamiliar e-wallet cluster, 2 a.m. sessions from a second device. Old world: fraud model says "legitimate"; credit score has not caught up. AURA: medium-scale shift, intent manifold = early bust-out / over-indebtedness transition. Action: consolidation offer + limit adjustment. **Loss prevented, customer saved.**

A synthetic identity ring files 200 flawless deepfaked applications. Individually pristine. Collectively: all 200 doppelgängers share one residual signature — identical application micro-cadence, one device cluster, synchronised behaviour templates. The Risk Field lights up as a cluster diverging from reality at once. **Ring killed at application, not at first missed payment.**

06 Governance: Why This Is Actually Deployable

The scary part of "continuously learning" risk models is regulatory: SR 11-7, EBA guidelines, and MAS TRM all demand stability, explainability, and version control. AURA resolves this with a dual architecture: the **AURA Core** — continuously learning, real-time, runs every decision through the latest embedding and the Arena's latest findings — alongside the **Regulatory Shadow**, a frozen, versioned, explainable gradient-boosted model running in parallel that produces the official, auditable score. The Core produces the decision; the Shadow produces the explanation. Conformal uncertainty is a first-class output: novel signatures route to humans automatically. Every score is explainable as a named divergence event, at a named timescale, with a named driver.

07 Business Model & Projected Impact

AURA is delivered as an enterprise platform with a hybrid commercial model: a base platform licence (annual, per-decision-volume tiered), plus a performance component indexed to verified loss reduction against a pre-deployment baseline. This aligns our economics with the bank's P&L outcome. The first 90-day shadow deployment is at zero licence cost.

FIGURE 3. Three-Year Projected Impact (Mid-Sized Bank, US\$50B Portfolio)

Metric	Year 1	Year 2	Year 3
Credit loss reduction	8–12%	15–20%	20–30%
Fraud loss reduction	20–25%	30–40%	40–50%
False-positive reduction	5–8%	10–15%	15–25%
Model deployment cycle (weeks)	12→6	6→3	3→1.5
Annual P&L; impact (US\$M)	15–25	35–55	50–80

08 Team

The founding team blends production risk-modelling experience with the research frontier in causal ML, generative twins, and reinforcement learning — the precise intersection required to deliver AURA's counterfactual replay loop and adversarial Arena in a regulated environment.

Role	Background
Dr. Maya Chen — CEO	Former CRO, DBS Consumer Bank. 18 years APAC retail credit. Led DBS model-risk transformation (2019–23).
Arjun Patel — CTO	Built fraud ML at Stripe (2017–22), Paytm (2014–17). Architected feature stores at 12M QPS. Stanford CS PhD candidate.
Dr. Lena Petrova — Chief Scientist	Co-author, foundational doubly-robust counterfactual estimation paper (JMLR 2021). Ex-Microsoft Research NYC.
Marcus Tan — Head of Regulatory Engineering	Ex-MAS regulator (TRM, 2015–20). Authored portions of the MAS model-risk notice. Ensures AURA ships audit-ready.

09 Implementation Roadmap

A four-phase, 36-month deployment minimises operational disruption. The first 90 days are zero-cost, zero-risk — the shadow deployment proves the embedding before any commercial commitment.

Phase	Timeline	Focus	Exit Criterion
1 — Shadow	0–6 mo	Build embedding + doppelgängers on existing features. Run as shadow.	Divergence baseline established; first counterfactual labels collected.
2 — Time Telescopes	6–12 mo	Deploy horizon heads. Live decisions on next-tx + 30-day heads in low-risk.	FP reduction $\geq 5\%$ on treated segments.
3 — Loop + Arena	12–24 mo	Counterfactual replay loop + Arena live. Train on interventions and exploits.	Credit loss $\downarrow \geq 10\%$ on treated portfolio.
4 — Genome + Action	24–36 mo	Risk Field primary substrate. Action Layer live. Migrate shadow to graph-explainable.	Full deployment; MRM sign-off.

10 The Ask

We seek three things from GXS Bank, in this order: (1) **A 90-day shadow deployment** at zero licence cost, on one product line (cards or unsecured lending), to establish the divergence baseline and prove the embedding; (2) **An executive sponsor** (CRO or CDO level) empowered to convene credit, fraud, and model-risk functions for joint review; and (3) **Access to three years of historical decision data** with outcome labels, so counterfactual training can begin on day one. In return, we commit to a measurable reduction in combined credit and fraud losses within twelve months — or AURA reverts to GXS Bank at no cost.